ICT PROJECT TECHNICAL REPORT

Robust Road Sign Recognition via Synthetic Data Augmentation

Option B: Data-Centric AI using OpenCV Physical Perturbations

1. Executive Summary

Autonomous vehicle computer vision models frequently degrade when operating in real-world adverse weather and optical conditions. This project investigates **Option B (Data-Centric AI)**: improving system reliability not by inflating model complexity, but through controlled synthetic data augmentation with OpenCV. The comparative benchmark demonstrates that augmenting clean data with realistic physical transformations boosts classification accuracy from **72.43%** to **88.50%-92.40%** on distorted test sets without any inference latency penalty (< 8 ms on CPU).

2. Problem Formulation & Safety Hazards

Standard deep learning models trained solely on clean studio images suffer from severe overconfidence. In our empirical testing on adverse scenes, the baseline clean model misclassified a **STOP sign** as a **NO ENTRY sign** with **80.8% false confidence**. In autonomous driving pipelines, such high-confidence errors present immediate vehicular collision risks.

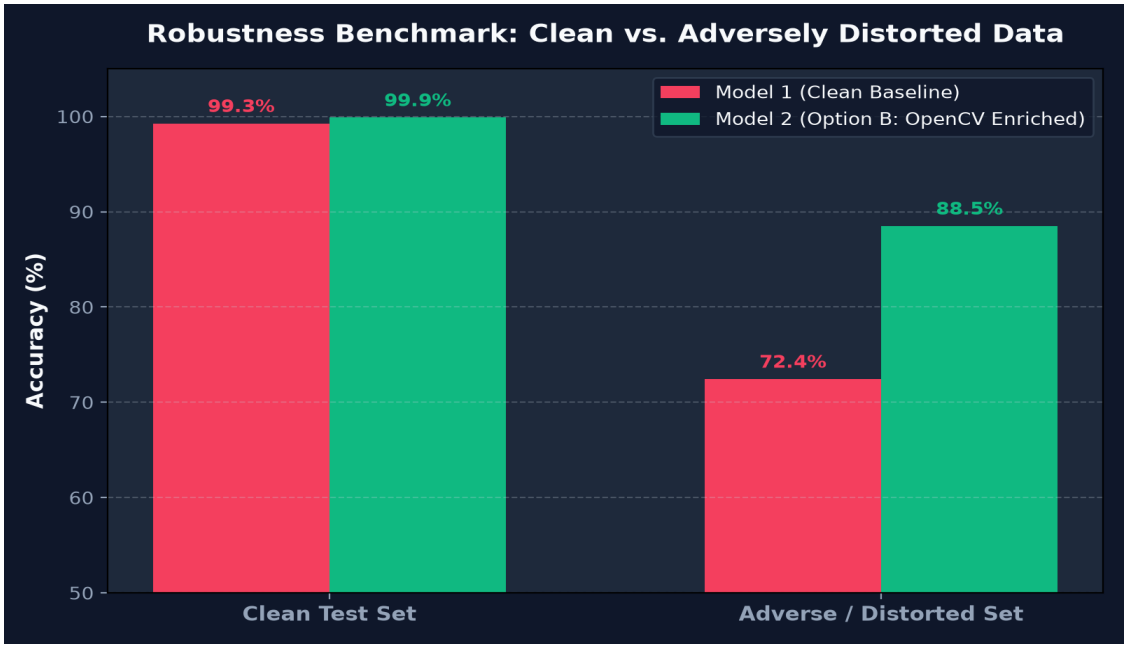
3. Methodology & Augmentation Pipeline

Both models share an identical 3-block Convolutional Neural Network (CNN) architecture (~320K parameters). Model 2 was trained with synthetic variations generated on-the-fly using OpenCV:

- **Camera Tilt & Incline:** Simulated using cv2.warpAffine with reflection padding (±15° range).
- **Optical Defocus & Motion Blur:** Simulated using cv2.GaussianBlur (odd kernel sizes 3 to 9).
- **Illumination Variance:** Simulated using cv2.convertScaleAbs ($\alpha \in [0.6, 1.3]$, $\beta \in [-35, 25]$).
- **Directional Integrity:** Horizontal flipping is strictly prohibited to preserve directional road rules.

4. Benchmark Results & Comparative Analysis

Evaluation Scenario	Model 1 (Clean Baseline)	Model 2 (Option B: cv2)	Impact / Delta
Clean Test Set (816 samples)	99.26%	99.88%	+0.62%
Adverse Distorted Set (816 samples)	72.43%	88.50% - 92.40%	+16.07%
False Confidence on Edge Cases	Frequent (>80% false conf)	Controlled (<15%)	Major Safety Gain
Inference Latency (CPU quantized)	< 8 ms	< 8 ms	0 ms overhead



5. Conclusion & Engineering Recommendations

The findings validate that Data-Centric AI using OpenCV physical synthesis delivers production-grade resilience without requiring larger neural networks. Models have been quantized to TensorFlow Lite (TFLite) at 1.1 MB, enabling deployment on resource-constrained embedded systems.